

Literature Review: Intention Recognition and Calibrated Assistance from Raw Observation

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This review situates MICA (Minecraft Intention-Aware Collaborative Agent) within five research threads: assistance games, goal and intent recognition, Minecraft foundation models, structured action prediction, and confidence calibration. Although each thread supplies an essential ingredient, none combines explicit calibrated belief tracking with perception-based evidence under raw observation; consequently, that combination defines the gap this project addresses.

1 Assistance Under Hidden Goals

The formal foundation of this work is the assistance game. Hadfield-Menell et al. [1] formalized value alignment as Cooperative Inverse Reinforcement Learning (CIRL), a two-player cooperative game in which both agents are rewarded according to the human’s objective, although only the human observes that objective. Because the robot must therefore infer the goal from behavior while simultaneously assisting, the authors prove that isolated optimality is suboptimal in this setting; thus, an effective assistant must manage its uncertainty rather than act on a point estimate. However, CIRL specifies the problem rather than a system.

Puig et al. [2] operationalized this framework as the Watch-And-Help benchmark, in which an agent first observes a human demonstration (Watch) and subsequently assists with the same goal in a novel environment (Help). Two findings from this benchmark directly motivate the present work. First, the cost of incorrect inference is negative rather than merely zero: a helper acting on a wrong goal repeatedly undid the human’s work, and goal conflicts arose in 40% of episodes. Second, the authors explicitly identify online goal inference—updating the estimate during assistance rather than fixing it after a single demonstration—as the principal open problem. Accordingly, NOPA [3] extended this line by maintaining a set of goal hypotheses that are updated online as the human acts, and by modulating assistance according to inference uncertainty. Most recently, AssistanceZero [4]

scaled assistance games to collaborative Minecraft building—the same task family as MICA—using a Monte Carlo tree search policy whose goal inference is amortized into a neural prediction head.

Collectively, this thread establishes the problem structure, the cost of erroneous inference, and the value of explicit uncertainty. Nevertheless, all of these systems share one simplification: the assistant reads privileged, ground-truth environment state, whether as symbolic scene graphs [2, 3] or as oracle game state [4]. Moreover, none of them evaluates whether the stated confidence of the inference is calibrated, and assistance is assessed by task outcome alone. In contrast, a physically embodied assistant receives only noisy, partial perception; therefore, it must additionally determine when its own inference should not be trusted.

2 Goal and Intent Recognition

Goal recognition constitutes an older and broader tradition. Classical plan recognition [5, 6] infers goals by identifying the candidate whose optimal plan best explains the observed actions; however, its recognized weakness is that humans rarely behave optimally. Two responses to this weakness shaped MICA directly. First, GRAIL [7] replaces planning with learning by training one goal-conditioned policy per candidate goal via imitation learning from potentially suboptimal demonstrations, so that scoring a partial trajectory requires only a single forward pass; MICA’s learned likelihood heads follow the same principle, since they learn what pursuing each goal looks like from recorded sessions rather than assuming it. Second, Zhi-Xuan et al. [8] model the human as a boundedly rational, resource-limited replanner within an online recursive Bayesian filter, and demonstrate that modeling the planning process—rather than merely action noise—is what makes goal posteriors accurate on suboptimal and backtracking trajectories. Consequently, their latent internal-process variable is the direct ancestor of MICA’s reasoning-mode variable z , which distinguishes deliberate from habitual execution.

In parallel, a robotics line based on partially observable Markov decision processes [9, 10, 11, 12] established the pattern of a belief over human intent governing the degree of machine autonomy in shared-control settings. MICA’s escalating authority ladder (observe, suggest, preview, place) renders this principle explicit as a safety gate.

Finally, CLIPS [13] resolves a design question that MICA’s evaluation poses directly, namely where a large language model belongs within intent inference. In CLIPS, the language model serves exclusively as a likelihood term inside a Bayesian posterior and never as the decision-maker; notably, their direct-LLM baseline, in which GPT-4V names the goal outright, was the worst method evaluated, whereas the LLM embedded in the Bayesian frame performed best. MICA’s four-arm comparison mirrors this contrast, since the LLM-reader arm (0.292) underperforms the learned Bayesian tracker (0.375) on never-seen sessions. Furthermore, CLIPS handles the assistant’s own actions through a do-operator, and MICA pins the same rule: the agent’s own placed blocks are never treated as evidence about the human.

3 Minecraft Foundation Models Read in Reverse

Minecraft became a viable research platform because two works equipped it with foundation models. MineDojo [14] contributed an internet-scale knowledge base together with MineCLIP, a contrastive video–language model that scores how well recent video matches a natural-language task description. Meanwhile, VPT [15] trained a behavioral prior on approximately 70,000 hours of YouTube gameplay, pseudo-labeled by an inverse dynamics model. An ecosystem of acting agents subsequently emerged: STEVE-1 [16] follows text instructions by writing commands into MineCLIP’s latent space, multi-agent benchmarks [17, 18, 19] coordinate several builders, and BEDD [20] benchmarks fuzzy, human-judged tasks.

MICA’s relationship to this ecosystem is an inversion. Whereas each of these models was constructed to make an agent act, MICA executes the direction in reverse: VPT’s trunk serves as a frozen behavior encoder that summarizes what the human is doing, and where STEVE-1 writes a command into MineCLIP’s goal space to produce behavior, MICA reads from the same space by scoring observed behavior against each candidate goal expressed as text. No backbone is fine-tuned; instead, only lightweight task-specific heads are trained on the project’s own recorded sessions, because the dataset of 37 real building sessions is far too small to train models of this scale without memorization. The same pattern extends to three dimensions, since Uni3D [21], a pre-trained point-cloud encoder, reads the reconstructed structure as a single shape vector, following the structured-3D regime established by works such as SpatialLM [22].

4 Structured Action Prediction

MICA’s anticipation result depends on treating builder actions as a small formal language of 87 tokens—five action verbs together with coordinate and block-type arguments—and on training a decoder to predict what follows. Two works license this architecture. FastSceneScript [23] demonstrates that structured scenes can be decoded as token sequences using multi-token prediction with confidence-guided decoding, which is the same structured-language framing MICA applies to building actions. Moreover, Gloeckle et al. [24] provide the peer-reviewed basis for the decoder’s multi-token form by showing that predicting K future tokens through parallel heads on a shared trunk is both viable and beneficial. Their appendix also reports that attaching a multi-token loss to a next-token-pretrained model at fine-tuning time failed; consequently, MICA’s staged training recipe, which performs next-token pretraining on real sessions before the multi-token stage, was designed to respect rather than ignore this negative result.

5 Confidence Calibration

The final thread is the source of this project’s principal negative finding. Guo et al. [25] demonstrated that modern neural networks are systematically overconfident, in that their stated probabilities exceed their actual accuracies, and they standardized the measurement protocol of reliability diagrams and expected calibration error (ECE). MICA adopted this protocol as a pre-registered evaluation of its own belief, and the predicted failure was confirmed: above a stated confidence of 0.4, held-out reliability inverts. Therefore, the confidence-licensed placement route is disabled by its own threshold, which constitutes the project’s central safety mechanism. Notably, prior assistance systems [2, 3, 4] never performed this audit; in contrast, MICA treats calibration as a gate that operational authority must pass before any autonomous action is permitted.

6 Positioning

In summary, each neighboring line of work omits precisely one element that MICA supplies. The assistance-game lineage [1, 2, 3, 4] establishes the problem yet reads oracle state and never audits calibration; the goal-recognition lineage [7, 8, 13] contributes learned likelihoods, bounded rationality, and Bayesian discipline, yet operates in symbolic domains without perception; the Minecraft foundation models [15, 14, 16] supply powerful representations, yet were built for control rather than interpretation; and calibration research [25] defines the measurement, yet has not been applied to an assistant’s license to act. MICA therefore occupies the intersection: an explicit, recursive Bayesian belief over goal and reasoning mode [8, 13], whose likelihoods are learned from recordings [7], over evidence produced by frozen foundation encoders run in reverse [15, 14, 21], in the same task family as scaled assistance games [4]—together with the discipline none of these systems carried: the belief’s confidence is calibration-audited, and the agent’s authority to act is gated on passing that audit, which it has not yet passed. The literature thus supplies every ingredient; the contribution is their combination under robot-like observational conditions, evaluated by pre-registered criteria, with failures reported alongside successes.

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